Kiyan Amirian

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Available Oct 2026-Aug 2027 (flexible for later end dates) | U.S. Permanent Resident

Research Summary

PhD researcher combining atomistic simulation and machine learning for materials discovery. Trains and fine-tunes machine-learning interatomic potentials and designs diagnostic benchmarks of their reliability. Has hands-on experience in LLM inference, evaluation and quantization. Builds open-source research software and automated workflows, using coding agents for development, analysis and literature review.

Education

University of Maryland — **College Park, MD, USA**
Ph.D. in Materials Science and Engineering, GPA 4.0/4.0 — *2023-Expected Aug 2027*

Shiraz University — **Shiraz, Fars, Iran**
B.Sc. in Materials Science and Engineering, GPA 17.15/20 (3.5/4.0) — *2018-2023*

Research Experience

Graduate Researcher, UMD (Advisor: Prof. Yifei Mo; Collaborator: Prof. Haizhao Yang) *2024-Present*

FP-DeErr: error decomposition for foundation potentials

- Developed an **open-source framework** that decomposes foundation-potential (FP) errors by the physical quantities, configurations and workflow stages governing each task. Built its evaluation code and public leaderboard, which benchmarks 18+ FPs against DFT (ASE, MatCalc, pymatgen).
- Designed and built a configurable **benchmark automation pipeline** integrating model-specific environments and checkpoints with parallel SLURM job generation, result validation and leaderboard export. Released DFT-NEB reference data and chalcogenide benchmark subsets on Figshare.
- Designed new diagnostic metrics based on **force magnitude and angle** for highly accurate predictions, large-force-error atoms and far-from-equilibrium atoms (MatPES, OMat24), revealing failure modes obscured by average MAE/RMSE (force-error GitHub, leaderboard).
- Designed new **convex-hull agreement metrics** (ground state, within-phase and global hull minimum) and assessed elemental-ordering rankings using Top-1, Recall@k and Spearman's ρ on chalcogenide phase-change materials, separating intrinsic energy-ranking errors from FP-relaxation errors (phase-stability GitHub, leaderboard).
- Developed an automated workflow **benchmarking FP-NEB against DFT-NEB** on 154 Li- and Na-ion pathways, with new diagnostic metrics for convergence, profile shape and endpoint energy ranking beyond barrier accuracy, tracing barrier errors to intrinsic PES error, endpoint relaxation and along-path forces (ion-migration GitHub, leaderboard).

Machine-learning interatomic potential (MLIP) training for ion transport

- Designed and compared **training-data selection strategies** for fine-tuning MACE on DFT NEB data; the best strategy reduced migration-barrier MAE against DFT from 0.088 eV (zero-shot MACE-MPA-0) to 0.013 eV across hop types represented in training (GitHub).
- Training and fine-tuning MLIPs (MACE, CHGNet) to improve the description of ion transport for Li- and Na-ion conductor discovery. Benchmarking MLIP-MD diffusivities against AIMD in preparation for large-scale, more accurate simulations of **diffusivity and conductivity** (ongoing).

Computational materials discovery

- Designed a **screening workflow** to discover new Na-ion conductors. Identified K-containing compounds with structure types matching known Na-ion conductors. Replaced K with Na in selected frameworks to test whether the smaller Na-ion migrates more easily, using MLIP-NEB and DFT (ongoing).

Undergraduate Researcher, Shiraz University (Advisor: Prof. Ramin Ebrahimi) *2019-2022*

Continuum modeling of functionally graded steel

- Developed an algorithm to **predict tensile instability** in functionally graded low-carbon steel and compared its predictions with published experimental data (paper).

Selected Machine Learning Projects

KV-cache optimization for large language models *Fall 2025*

- Led a team comparing KV-cache compression techniques (StreamingLLM, KIVI, ZipCache) for long-context LLM inference. Reimplemented **KIVI 2-bit quantization** for LLaMA-2 7B/13B using PyTorch on CUDA GPUs (GitHub).
- Designed **evaluation workflows** across summarization, reasoning, and QA benchmarks, adding metrics for token match rate, entity flag rate, and sentence compliance.

Image classification and transfer learning *Spring 2025*

- Implemented logistic regression from scratch in NumPy; used PCA with an RBF SVM for MNIST (97.1% test accuracy) and fine-tuned VGG19 for a 10-class monkey-species dataset.

Face recognition and kernel methods *Spring 2025*

- Implemented a custom gradient-descent kernel SVM to address CVXOPT numerical failures with high-degree polynomial kernels. Benchmarked PCA/MDA with Bayes, k-NN, SVM, and AdaBoost on a 200-subject dataset.

Publications and Manuscripts

***K. Amirian**, R. S. Saravanan, F. Adams, C. E. Schwarz, Y. Mo* *arXiv, 2026*

FP-DeErr: Application-Oriented Error Decomposition for Foundation Potentials
arXiv:2609.05714 · GitHub · Leaderboard

*R. S. Saravanan, C. E. Schwarz, F. Adams, **K. Amirian**, Y. Mo* *Accepted, 2026*

A Simple Descriptor for Solid-State Ionic Conductors Across Chemistries
Advanced Materials

***K. Amirian**, Z. Abbasi, R. Ebrahimi* *2023*

Analytical and numerical instability analysis of functionally graded low-carbon steel. International Journal of Iron and Steel Society of Iran, 19(2), 66–72. doi:10.22034/IJISSI.2023.1990580.1262

Skills

Programming: Python (NumPy, pandas, SciPy, Matplotlib), MATLAB, Fortran
Atomistic Methods: DFT, NEB, molecular dynamics
Simulation Codes: VASP, LAMMPS
Materials Analysis Tools: ASE, pymatgen, MatCalc
MLIPs & Foundation Potentials: MACE, CHGNet, M3GNet, UMA, Orb, MatterSim, DPA4, eSEN
ML Frameworks: PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn
ML Methods: MLIP fine-tuning, LLM KV-cache quantization, transfer learning, CNNs, SVMs
Continuum Modeling: Abaqus (FEA)
Coding Agents: Claude Code, Codex
Computing & Tools: HPC (SLURM), Bash scripting, Git, LaTeX

Honors & Awards

Dean's Fellowship Award, University of Maryland, $43,000 *Spring 2023*
Ranked 2nd of 56, B.Sc. Materials Science and Engineering, Shiraz University *Spring 2022*

Relevant Graduate Courses

Introduction to Atomistic Modeling (4.0/4.0)
Natural Language Processing (4.0/4.0)
Statistical Pattern Recognition (4.0/4.0)
Machine Learning for Materials Science (4.0/4.0)

Teaching Experience

TA: Kinetics, Diffusion and Phase Transformations (Fall 2024)
TA: Mechanical Properties II (Spring 2022)
TA: Fundamentals of Computer (Spring 2020)
High-school algebra instructor (Fall 2019)

References

Yifei Mo Professor, MSE, UMD yfmo@umd.edu
Haizhao Yang Professor, Math, UMD hzyang@umd.edu